

Unit 5, Lesson 3: Interpreting Equations



Benchmarks

MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



Learning Targets

- ☐ I can define variables in a real-world context.
- ☐ I can interpret equations and their parts in a given context.



5.3.1 Warm-Up: Orange Juice

1. A typical orange produces 86 grams of juice. The juice contains 9 grams of carbohydrates.
 - a. If 1 gram of carbohydrate contains 4 calories, how many calories come from the carbohydrates in one orange?
 - b. The total number of calories of the juice from one orange is 39. What percentage of those calories comes from the carbohydrates in the juice?
 - c. Carbohydrates include sugar. The sugar that occurs in orange juice is natural sugar. If a typical orange contains 7 grams of sugar, what percentage of the calories from carbohydrates comes from sugar? Round to the nearest tenth.



5.3.2 Guided Instruction: What Is a Parameter?

1. In the expression $4x$, the 4 is the coefficient and the x is the variable. The expression $4x$ could be used to represent the number of calories in orange juice that comes from carbohydrates since 1 gram of carbohydrate contains 4 calories.



A **parameter** is a quantity that influences the output.

In the expression $4x$, what is the parameter?

2. The cost of an item is \$8.50.
 - a. Write an expression that can be used to determine the cost of x items.
 - b. Discuss with your partner why \$8.50 is a parameter. Summarize your discussion.

3. The price of an item is P . During a sale, the price is discounted by 20%, so the new price is $0.8P$. This could also be expressed as $(1 - 0.20)P$.

Complete the table.

Expression	Parameter
$0.8P$	
$(1 - 0.20)P$	

4. Discuss with your partner which values in the U-Pick Blueberries sign are parameters. Justify your choice.

U-Pick Blueberries

\$4 per pound

\$1 parking



5. The average price, P , in dollars, of a 16 ounce bag of potato chips, x , since 1980, can be modeled by the function $P(x) = 0.07x + 2.20$.
 - a. Which value in the function is a parameter?

- b. What does the value 2.20 mean in terms of the price of potato chips?
- c. What does the value 0.07 mean in terms of the price of potato chips?

6. The students in the Visual and Performing Arts program at Booker High School in Sarasota County, Florida are planning a field trip to the Ringling Art Museum. The cost of admission is \$5 and the cost of renting a bus is \$115. There will be 9 chaperones. The budget for the field trip is \$600.

- a. Work with your partner to determine which values, if any, are parameters. Write a justification for each.

Value	Parameter?	Justification
5	☐ yes ☐ no	
9	☐ yes ☐ no	

115	☐ yes ☐ no	
600	☐ yes ☐ no	

- b. Use the values to complete the equation that can be used to find the number of students, x , who can go on the field trip.

$$\underline{\hspace{1cm}}(x + \underline{\hspace{1cm}}) + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$



5.3.3 Activity: Card Sort

1. Work with your partner to sort the cards provided by your teacher.
 - a. Record the two lowercase letters on the card that matches the meaning of each value.

Real-World Context	Meaning of Each Value	
1	1991	
	6.88	
	0.37	
	3	
2	2010	
	19.116	
	1.352	
	43.452	
3	8	
	0.28	
	34.60	
	2.40	

- b. Record the letter and the equation that matches the context for each scenario.

Real-World Context	Equation
1	
2	
3	



5.3.4 Wrap-Up: Growth Mindset

1. Mistakes are an opportunity to learn. Complete the statement to highlight a mistake you learned from.

My favorite mistake from today was ...

Means
I
Start
To
Acquire
Knowledge
Experience
Skills